S1 Text. Supplementary Notes

These notes collect material that supports the main text without being part of its argument. They first give the channel model and the conjectured count law in the form used below. They then treat counts more or less regular than Poisson counts. For the Conway-Maxwell-Poisson count of the main text they prove that the optimal input has finitely many rates. They also treat a spontaneous rate, codes for windows that hold few spikes, a test for discrete levels on six public datasets, and where a new rate appears. Results here carry the same status words as in the main text.

S1.1 The neuron as a channel

The input $x \in [0, A]$ is the expected count produced by the stimulus in one window, the output Y is the count, and $Y \sim \text{Poisson}(x+\lambda)$, where $\lambda \geq 0$ is the baseline count per window (the dark current of the optics literature). For an input law P with output mixture Q, let $D(x) = D\big(\text{Poisson}(x+\lambda) \,\|\, Q\big)$. Then P achieves the capacity C if and only if

$$D(x) \leq C \quad \text{on } [0, A], \quad \text{with equality on } \operatorname{supp} P, \tag{1}$$

and the optimal P is unique and has finitely many atoms [1,2]. We write $N(A, \lambda)$ for their number and $N(A) = N(A, 0)$. The Fisher information of the Poisson count is $1/(x + \lambda)$, so in the Fisher coordinate $t = 2\sqrt{x + \lambda}$ the noise has unit size everywhere: two rates one unit apart in t are equally hard to tell apart wherever they lie. The input range has half-length

$$h(A, \lambda) = \sqrt{A + \lambda} - \sqrt{\lambda}. \tag{2}$$

The reference is the Gaussian channel $Y = X + Z$ with $Z \sim N(0, 1)$ and $X \in [-A_g, A_g]$, whose input is discrete at the optimum [3], whose Fisher coordinate is its input itself and whose half-length is A_g. Read in the Fisher coordinate, the Poisson channel approaches this Gaussian channel at large counts [4], and Jeffreys' prior, uniform in that coordinate, is asymptotically capacity-achieving [5]. We write $N_G(h)$ for its number of levels at half-length h, a staircase in h, and $A_g(K)$ for the half-length at which its K-th level appears.

Two families describe neurons that are not Poisson. A count more regular than Poisson has Fano factor below one; the binomial count is the simplest case, with Fano factor $1 - p$. A count more variable than Poisson has Fano factor above one, the negative binomial being the standard case, with Fano factor $1 + \text{mean}/\rho$ for size parameter ρ. The Conway–Maxwell–Poisson (COM-Poisson) family [6] covers both directions with one shape parameter ζ, sub-Poisson for $\zeta > 1$ and super-Poisson for $\zeta < 1$, and contains the Poisson count at $\zeta = 1$; it is a standard model of spike counts [7]. Refractoriness can also be modelled as spike history, by a generalized linear model (GLM) whose conditional intensity is multiplied by $\exp(\sum_j \eta_j y_{t-j})$, where y_{t-j} marks a spike j bins earlier and $\eta(s) = -a\, e^{-s/\tau}$ at lag s. We use $\tau = 4$ ms, amplitudes $a = 0$, 2, 4 and 8, and bins of 2 ms, and the history is reset at the start of each window. A bin holds at most one spike, so even the kernel $a = 0$ gives a count with Fano factor 0.89 at a rate of 60 Hz in 50 ms windows; the GLM without history is therefore not the Poisson neuron.

In a short decision the reader does not see many windows. Suppose a message must be decoded from n windows (the blocklength) with error probability ϵ. In the normal approximation, which drops terms of order $\log n$, the largest number of messages M^* satisfies

$$\log M^*(n, \epsilon) \approx nC - \sqrt{nV}\, Q^{-1}(\epsilon), \tag{3}$$

Table 1: Symbols

symbol	meaning
x, $A = LT$	expected count in a window; peak count
λ	baseline count per window
$N(A, \lambda)$, $N_G(h)$	number of rates (Poisson); of levels (Gaussian)
$h = \sqrt{A+\lambda} - \sqrt{\lambda}$	Fisher half-length of the rate range
A_K, $A_g(K)$	peak count (Poisson) and half-length (Gaussian) at which level K appears
$c(K)$	wall constant, $\sqrt{A_K} - A_g(K)$
n, ϵ, V	number of windows, error probability, dispersion

where V is the channel dispersion, the variance of the information density, and Q^{-1} the inverse of the Gaussian tail function [8]. With an energy cost the same form holds with the cost-constrained C and V [9]. We charge energy linearly in the mean rate, as the Attwell–Laughlin budget does [10]. Table 1 collects the symbols.

S1.2 Long windows: how many rates

S1.2.1 The count law

In the Fisher coordinate the levels of the optimal Poisson code are spaced like the levels of the unit-noise Gaussian channel, except near silence. A silent neuron never spikes, so the output at $x = 0$ is degenerate, and it sits at a finite Fisher distance from the rest of the range. In the computed optima the levels next to silence form a boundary layer, and the law below treats its effect as an extra length that changes slowly with K. For the Gaussian channel, Wang, Barletta and Dytso [11] summarize an argument of Zhang [12] that the optimal input approaches Jeffreys' prior in the bulk, with a discrete pattern next to the boundary. Abbott and Machta found the first levels next to an endpoint at fixed Fisher distances [13]. We conjecture that the Poisson count is the Gaussian count read at a shortened half-length,

$$N(A, \lambda) = N_G\big(h(A, \lambda) - c(K, \lambda)\big), \tag{4}$$

where $c(K)$ is the wall constant. Since K is the count being predicted, (4) is implicit, and we evaluate it at the count changes A_K, where K is known. We call silence a *wall* because the constraint $x \geq 0$ is active there: the level at silence carries no derivative condition in (1). The constant is defined from the measured count changes as $c(K) = \sqrt{A_K} - A_g(K)$. It can also be predicted without them, from a half-line chain. The chain has one level fixed at silence with a free weight, thirty free levels, and ten frozen levels of unit weight at spacing g in the Fisher coordinate. That makes 41 levels in all, more than the full problem has for $K \leq 24$. The chain's equations ask that D take the same value at every free level and at the first frozen one, and that $D' = 0$ at the free levels: 61 equations for the 30 positions and 31 weights. We solve them once with the Poisson kernel and once with the Gaussian kernel, and define the *excess* $E(g)$ as the Fisher position of the thirtieth free level in the Poisson chain minus its position in the Gaussian chain. We call the resulting prediction the *closure*:

$$\begin{aligned} c(K, \lambda) &= \tfrac{1}{2} E\big(g_{\text{eff}}(K), \lambda\big), \\ g_{\text{eff}}(K) &= A_g(K+1) - A_g(K-1), \end{aligned} \tag{5}$$

where the factor $\frac{1}{2}$ converts the Fisher coordinate $t = 2\sqrt{x}$ to the scale of the half-length $\sqrt{A}$, and g_{eff} is the local spacing of the Gaussian levels at the count change. The chain law $E(g) = 0.9840 - 0.2347\,g$ is a fit to the chain's own excess on $1.43 \le g \le 2.35$ (root-mean-square deviation 0.0012); it never sees a count change. The estimator g_{eff} was chosen among three by comparing them with the measured wall constants; the other two, the one-sided spacing $2(A_g(K+1) - A_g(K))$ and the mean spacing $2A_g(K)/(K-1)$, deviate by 0.0043 and 0.0313 against 0.0015 over $K = 5$ to 24.

A worked case shows how the law is used. For $K = 15$, Table 2 gives $A_g(14) = 12.0413$ and $A_g(16) = 13.5819$, so $g_{\text{eff}} = 1.5406$, $E = 0.6224$ and $c = 0.3112$. Adding c to $A_g(15) = 12.8187$ gives $\sqrt{A_{15}} = 13.1299$, against the measured 13.1302: the fifteenth rate of the optimal code appears at a peak count of 172.40. Nothing in that line was fitted to a Poisson count change.

S1.2.2 The evidence

Table 2 lists the count changes for $K = 5$ to 24. The wall constant rises from 0.237 to 0.335 with steps that shrink, and the closure (5) matches it with a root-mean-square deviation of 0.0014 over $K = 5$ to 20 and 0.0015 over $K = 5$ to 24. The residual follows a smooth trend. It rises steadily from -0.0026 at $K = 6$ to $+0.0019$ at $K = 24$ and changes sign near $K = 14$.

Four count changes were registered before they were computed, each with a tolerance of ± 0.15 in A. The twenty-first was registered at $A = 307.99$ from an earlier local fit of the closure; the full-range law above had been recorded before the computation, and it gives 308.03, a number first written down after it. The twenty-second was registered at 333.14 by the full-range law. The computation returned 308.09 and 333.20. Both residuals, about $+0.0016$ in c, matched the size of a finite-length correction, $+0.002$, that had been estimated beforehand from chains of twenty, thirty and forty free levels, and only then did we adopt it. With the correction the law placed the next two at 358.97 and 385.39, and without it at 358.90 and 385.31; both sets were registered, and the computation returned 358.97 and 385.38 (Fig. 1). Its predictions at 25, 47 and 67 rates are reported in the count-law section of the main text.

S1.3 Neurons more regular or more variable than Poisson

A neuron more regular or more variable than Poisson keeps the wall at silence. The binomial and negative-binomial counts both have a degenerate output at silence, at a finite Fisher distance. The first two *gaps* next to silence, the spacings between neighboring rates in the Fisher coordinate [14], are the Poisson ones: 3.126 and 2.260 Fisher units for the binomial with $n_b \ge 1000$ trials, and 3.125 and 2.260 for the negative binomial with size $\rho = 1000$. These place the first two levels above silence at 3.13 and 5.39; Abbott and Machta found 3.13 and 5.41 for the Poisson count [13]. Unlike the Poisson count, whose variance equals its mean, these counts have a variance-to-mean ratio that changes with the rate, and that adds one term. Writing $q(x) = \sqrt{\text{mean}/\text{variance}}$ and $q_0 = q(0)$, the count is read at the effective half-length

$$h_{\text{eff}} = h_F - \kappa\,\frac{h_F - q_0\sqrt{A}}{K}, \tag{6}$$

where h_F is the Fisher half-length, A the peak count and $\kappa = 0.42 \pm 0.02$ a measured constant. It comes from 54 count changes of the binomial count on $[0, \varrho n_b]$, a fraction ϱ of its full range with peak count $A = \varrho n_b$ and one wall ($K = 9$, 11, 17 and 18). Seven more come from the binomial channel on its full range, where both ends are walls ($K = 18$ to 24). Six count changes of the negative binomial give 0.39. The term vanishes for the Poisson count and for any count whose variance-to-mean ratio

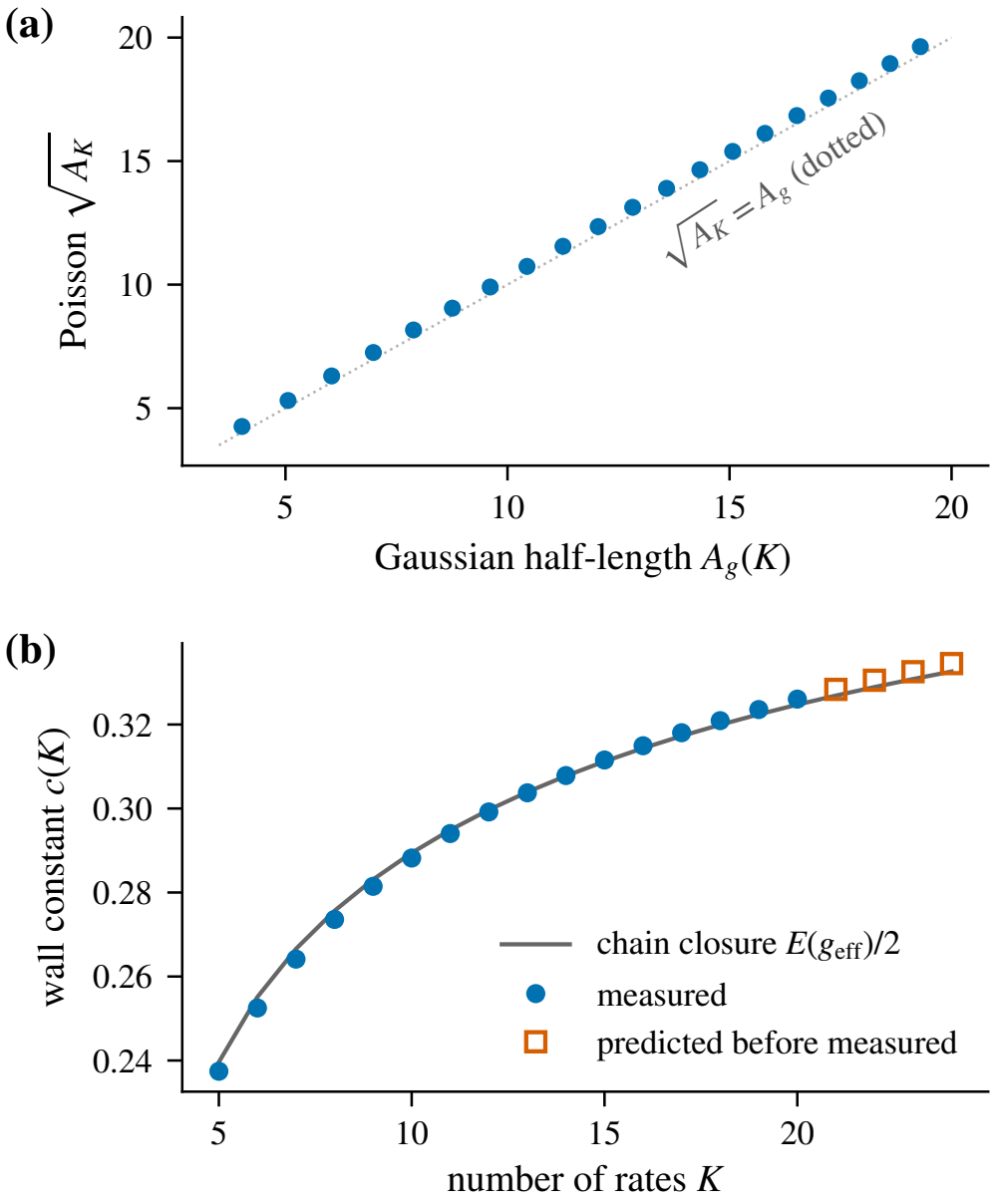


Figure 1: The Poisson count is the Gaussian count at a shortened Fisher half-length. (a) Poisson count changes $\sqrt{A_K}$ against the Gaussian ones $A_g(K)$, $K = 5$ to 24: a line of slope one, offset by the wall constant. (b) The wall constant $c(K)$, measured (dots) and predicted by the chain closure (line); open squares, $K = 21$ to 24, were predicted before they were computed.

Table 2: Count changes of the Poisson neuron and the wall constant. A_g: Gaussian half-length; A_K: Poisson peak count; $c = \sqrt{A_K} - A_g$; closure $E(g_{\text{eff}})/2$. Rows 21–24 were predicted before they were computed; the closure column omits the +0.002 correction used for the predictions of rows 23–24. Every A_g is relocated at 128 bits and every A_K comes from the newborn-weight unfolding at 40 digits.

K	$A_g(K)$	A_K	$c(K)$	closure	difference
5	4.02128	18.1366	0.2374	0.2396	−0.0022
6	5.05808	28.2022	0.2525	0.2551	−0.0026
7	6.03983	39.7401	0.2641	0.2666	−0.0024
8	6.97896	52.5995	0.2736	0.2756	−0.0021
9	7.88348	66.6665	0.2815	0.2831	−0.0016
10	8.75901	81.8521	0.2882	0.2894	−0.0012
11	9.60969	98.0839	0.2940	0.2949	−0.0008
12	10.43871	115.3023	0.2992	0.2997	−0.0005
13	11.24859	133.4562	0.3037	0.3039	−0.0002
14	12.04135	152.5025	0.3078	0.3078	+0.0001
15	12.81867	172.4026	0.3115	0.3112	+0.0003
16	13.58195	193.1236	0.3149	0.3144	+0.0006
17	14.33238	214.6348	0.3180	0.3173	+0.0008
18	15.07098	236.9105	0.3209	0.3199	+0.0010
19	15.79863	259.9251	0.3236	0.3224	+0.0011
20	16.51609	283.6578	0.3261	0.3247	+0.0013
21	17.22406	308.0874	0.3284	0.3269	+0.0015
22	17.92312	333.1959	0.3305	0.3289	+0.0016
23	18.61381	358.9655	0.3326	0.3308	+0.0018
24	19.29662	385.3810	0.3345	0.3326	+0.0019

is constant, as the generalized Poisson count of Consul confirms ($|\Delta h| < 0.0005$). It also vanishes as $K \to \infty$, so the Fisher-length law is its large-K limit. Registered before they were computed, (6) placed the 25th and 26th count changes of the full-range binomial channel at $n_b = 174.94$ (with $\kappa = 0.435$, the value measured at the time) and 186.36 (with $\kappa = 0.42$). The registered ranges for κ from 0.40 to 0.46 were 174.76–175.07 and 186.24–186.55, and the solver returned 174.934 and 186.386. At $\kappa = 0.42$ the two predictions are low by 0.07 and 0.04. For the binomial channel, the published bounds on the count and the computed optimal inputs are in [15–17].

The COM-Poisson neuron also has a degenerate output at silence, so the same construction predicts its count, with its own Fisher length and dispersion ratio; we have not computed its wall constant. What we did compute is its count at Fano factors 0.9, 1.13, 1.3 and 1.5, which bracket the median Fano factors of the datasets of Section S1.6 (0.90 to 1.12, and 1.39 in V1). For $A \leq 12$ it differs from the Poisson count by at most one rate, and variability above Poisson never raises it. The auditory-nerve fibres of the main text, with a median Fano factor of 0.355, lie far below this range; for them the main text computes the number of rates for a COM-Poisson count matched to each fibre. In long windows, then, the number of rates is a property of the Fisher length of the rate range and of the type of the endpoint at silence. A neuron's Fano factor enters only through these and, at finite K, through the dispersion ratio at silence in (6).

S1.4 A baseline rate

A baseline rate removes the degenerate output: with $\lambda > 0$ a silent stimulus still produces spikes, and the endpoint at silence becomes regular. The wall constant then falls from its Poisson value

toward the Gaussian value, zero; at $K = 11$ it is 0.29 at $\lambda = 0$ and 0.02 at $\lambda = 1$. The chain with the same kernel predicts this crossover with no parameter, to 0.002 at $K = 11$ (six baselines) and to 0.001 at $K = 15$ (three). In the Fisher offset of the endpoint, $d = 2\sqrt{\lambda}$, the change is linear at small d, $E(g, 0) - E(g, \lambda) = 1.06\, d$, and the remaining excess decays as $0.055/d$ at large d; a baseline of 10^{-3} spikes per window already removes a tenth of the wall.

The law (4)–(5) with this chain predicts the twenty-five count changes in the published optimal inputs of Barletta and Dytso [18] for $\lambda = 1$, 10 and 100 (nine, eight and eight changes, from three rates up to eleven). The root-mean-square deviations are 0.005, 0.004 and 0.003 in the Fisher half-length, and the largest is 0.009. The six changes to three and four rates lie below $K = 5$, where g_{eff} exceeds the range on which the chain law was fitted; for them the agreement is with the law's extrapolation. The chain also gives the second rate of the code: its first gap g_1 next to silence places it at an expected count of $(g_1/2 + \sqrt{\lambda})^2 - \lambda$ above the baseline. This is 4.64, 10.48 and 29.06, against the published 4.64, 10.49 and 29.06 at the largest peak count of each run. For every baseline the lowest rate of the optimal code stays at silence, where the neuron fires at the baseline rate. Where a new rate appears is also set at silence. An inserted rate appears at 0.62 of its gap from the silent side, and a split happens at 0.12 of the gap next to the parent rate, against $\frac{1}{2}$ and 0 for the Gaussian channel. The baseline carries both values to the Gaussian ones (Section S1.8).

S1.5 Short windows

S1.5.1 A window that holds few spikes

Spike counts are commonly computed in windows of about 50 ms [19], and such a window often holds only a few spikes. There the asymptotic capacity overstates what a neuron delivers; Kostal and Shinomoto found a drop of about an order of magnitude under finite coding [20]. By (3), the relative loss $\phi = \sqrt{nV}\, Q^{-1}(\epsilon)/(nC)$ of a Poisson neuron depends mainly on the expected number of spikes N_{sp} in the whole decision, $\phi \approx Q^{-1}(\epsilon)/\sqrt{N_{\text{sp}}}$. The approximation holds when C^2/V is close to the mean count per window, since $\phi = Q^{-1}(\epsilon)\sqrt{V/(nC^2)}$. In the windows used here C^2/V is about 12% above the mean count (at $A = 3$, $C = 0.5944$ and $V = 0.2224$ nats, so $C^2/V = 1.59$ against a mean count of 1.41). In continuous time with a slack power constraint the two are equal [21]; when an energy budget binds they are not, and the approximation fails. At $\epsilon = 0.01$, where $Q^{-1}(\epsilon) = 2.33$, a decision built on fifty spikes loses a third of the asymptotic rate. With fewer than about sixty spikes ϕ exceeds 0.3, where the expansion is no longer accurate and overstates the loss, but the loss stays large. Below a total of about $Q^{-1}(\epsilon)^2 = 5.4$ spikes the right side of (3) is negative, that is, less than one message can be decoded reliably, a known consequence of the expansion.

S1.5.2 The best short-window code is on-off

The loss also changes which code is best. Bethge, Rotermund and Pawelzik found binary tuning optimal for rate coding at high temporal resolution [22]; here the objective is the right side of (3), and the comparison is with the exact long-window optimum. We maximized that objective by search over input laws on a grid of rates from 0 to 60 Hz, for the Poisson neuron and for the four GLM kernels of Section S1.1 (induced Fano factors 0.89 to 0.46). There were 130 cases over energy budgets and blocklengths $n = 10$ and 50 at $A = 3$, in windows of 50 ms. Another 55 were at a fixed decision time of 500 ms, with windows of 10 to 100 ms, that is, peak counts from 0.6 to 6 and n from 50 down to 5; eleven of them were for the Poisson neuron. In all 185 cases the maximizer was binary: silence and one firing rate. A graded law was open to the search in every case and was

never chosen. Three limits apply. The maximizer was found by search and is not certified. The cases at $n = 10$ and all 55 at fixed decision time hold at most 30 expected spikes per decision, so $\phi \approx Q^{-1}(\epsilon)/\sqrt{N_{\mathrm{sp}}}$ is at least 0.4 there, outside the accuracy range of (3). And in many cases the long-window code is on-off as well, so they cannot show a contrast: at fixed decision time only the peak counts 3.5, 4.5 and 6 exceed the first count change of the Poisson neuron. An exact finite-blocklength bound answers this for small peak counts. The cases are the Poisson neuron without a spontaneous count at peak counts 4, 5 and 6, blocklengths 5, 10 and 20 and error probabilities 0.1 and 0.01. In each, the dependence-testing achievability bound [8] was computed with a certified bracket for 19 on-off laws, 45 graded laws and the capacity-achieving law. The best on-off law reached the larger number of messages in 16 of the 18 cases. A graded law did better only at peak count 6 with error probability 0.01 and blocklengths 10 and 20, by 0.196 and 0.130 nats. The on-off choice therefore survives outside the approximation at these peak counts and gives way first where the peak count is largest and the error smallest.

The long-window optimum behaves differently. Without an energy budget it is binary up to $A = 3.3679$ and graded beyond, so it changes at the first count change of Section S1.2.1, and beyond it the graded code soon carries much more than any on-off code. An on-off code carries at most $\log 2 = 0.693$ nats per window, while the capacity is at least 0.762 nats at $A = 6$, 0.916 at $A = 10$ and 1.106 at $A = 18$ (Blahut–Arimoto on 601 rates). An on-off code therefore loses at least 9%, 24% and 37% of the capacity there, and at $A = 4$ the best on-off code loses 1.1%. With an energy budget the code can be graded at smaller peak counts. At $A = 3$ we compared the Blahut–Arimoto optimum with every binary code on a 0.5 Hz grid and call a budget graded when the former exceeds the latter. For the neuron without spike history the long-window optimum is graded at budgets from 4.4 to 20.2 Hz (Fig. 2). Over the four kernels, 42 of 83 distinct budgets are graded in this sense. At the others the Blahut–Arimoto value falls up to 1.5×10^{-4} nats below the best binary code, so a graded optimum there is neither shown nor excluded. Where a graded code was found it gains at most 3.8×10^{-3} nats per window over the best binary code, 0.8% of the cost-constrained capacity at that budget. Near $A = 3$ the objective is nearly flat between graded and on-off codes, so a small dispersion term is enough to tip the choice to on-off. A prediction that rests on a nearly flat objective is fragile, and at $A = 3$ we report the effect as a change in which code is optimal; the information at stake there is small.

For a physiologist the switch is a statement about two numbers, the peak rate and the counting window. Without an energy budget the long-window code is graded only when $LT > 3.3679$, which at a 50 ms window needs a peak rate of 67 Hz. Driven layer-4 neurons of awake trained animals fire at a mean of 12 Hz, and at 28.5 Hz one standard deviation above that mean [23], below that, while high-rate cells or long counting windows can pass it. Without an energy budget, a neuron that cannot pass it uses the on-off code in long windows, and the short-window objective chose on-off as well.

S1.5.3 Refractoriness as spike history and as count regularity

A refractory kernel acts only after a spike. When the history is reset at the start of each window, the probability of a silent window is therefore e^{-xT} for every kernel, the void probability of a process with an empty history [24]; at $xT = 3$ we computed $0.04978707 = e^{-3}$ for all four kernels. With an on-off code any count above zero identifies the active rate, so the information depends on the kernel only through that probability. At $A = 3$, where the capacity-achieving code of each of the four GLM channels is on-off (computed), every GLM neuron therefore has exactly the Poisson capacity, 0.5944 nats per window. Above the first count change the Poisson optimum is graded and the identity says nothing. For their continuous-time channel, Kostal and Shinomoto conclude

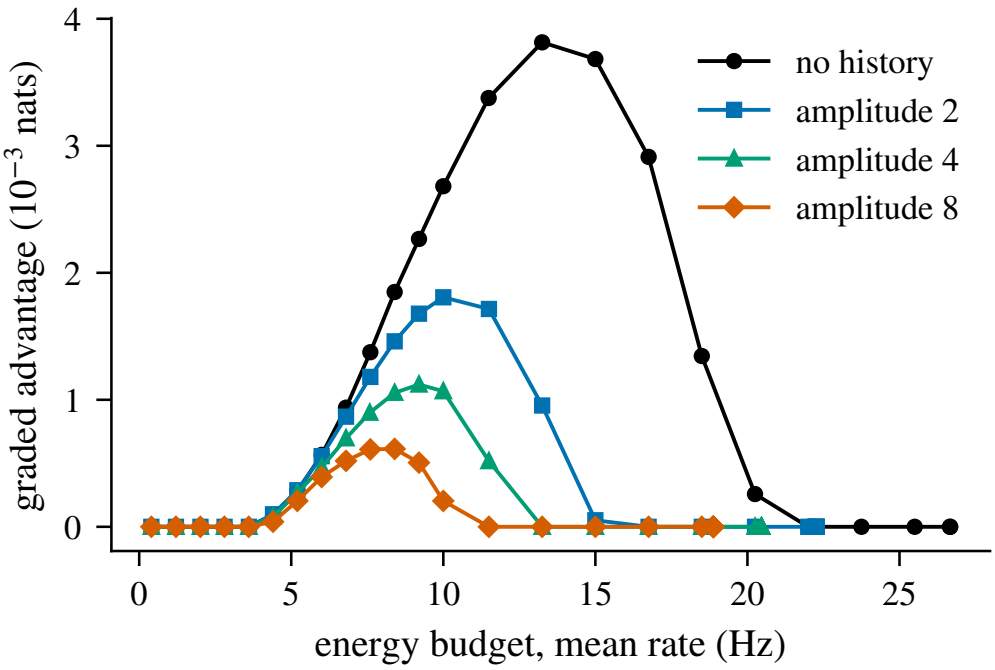


Figure 2: Under an energy budget the long-window code can be graded below the first count change, but it gains little. Peak count $A = 3$ (60 Hz, 50 ms windows): advantage of the Blahut–Arimoto optimum over the best binary code on a 0.5 Hz grid, for a neuron without spike history and refractory amplitudes $a = 2$, 4, 8 ($\tau = 4$ ms); zero means that no graded code was shown. The maximizer of the normal approximation was binary in every case.

more generally that a non-Poissonian neuron cannot increase the capacity through causal feedback, provided that the input satisfies their peak and average constraints and that the spontaneous activity is Poissonian [20]. The silent-window identity is the mechanism in our setting. A dead time at the detector is a different model: in the low-power regime it leaves the leading term of the discrete-time Poisson capacity unchanged unless it limits the output rate [25]. The capacity per unit cost of a refractory counting model is studied in [26]. Spike timing inside the window adds at most 2.7×10^{-4} nats per window over the count, so the count is nearly sufficient.

Refractoriness does lower the energy cost, because it suppresses spikes: the realised rate of the top symbol falls from 56.5 to 40.0 Hz as the kernel strengthens. At a matched mean rate between 3 and 25 Hz, in long windows, the GLM neuron with induced Fano factor 0.46 therefore beats the Poisson neuron, by up to 0.072 nats per window near 10 Hz and by 0.006 at 25 Hz. A COM-Poisson count of similar Fano factor (0.5) does better again, by 0.026 to 0.117 nats per window over the GLM neuron, because it also lowers the probability of a silent window, which the GLM leaves at its Poisson value. At Fano factor 0.5 and a mean count of 2.83 that probability is 0.0089, against 0.059 for a Poisson count of the same mean. Refractoriness as spike history and as count regularity therefore behave differently, and a count model of a refractory neuron overstates what refractoriness buys (Fig. 3(b)).

S1.5.4 The energy–capacity front

The energy–capacity front plots the best information per window against the mean firing rate it costs; Kostal and Shinomoto drew it for the Poisson neuron on the Attwell–Laughlin axis [20], and energy–information trade-offs of neurons go back further [27–29]. For long windows the front is the capacity–cost function, computed by cost-constrained Blahut–Arimoto, so each point is a global optimum. For short windows the front was found by search over input laws under a direct constraint on the mean rate. A Lagrangian sweep cannot replace the direct constraint here: the short-window objective is not concave in the input, and at $n = 10$ a sweep over 121 multipliers never reaches rates between 5.32 and 22.34 Hz. Fig. 3(a) shows both fronts for COM-Poisson neurons with Fano factors from 0.3 to 2; both rise smoothly with the energy budget. Lower Fano factor is better at every energy but one in each front. In short windows it is better at 43 of 44 comparable energies; the exception is Fano factors 1.5 and 2 at $n = 50$ and 3.6 Hz, which survived a tenfold finer search. In long windows the same pair is reversed at 0.4 Hz. The dispersion term thus does not reorder the neurons. The efficient operating point moves, however. The Attwell–Laughlin budget adds a static cost to the cost per spike. On it, the information per ATP of the Poisson neuron in long windows is largest at 1.2 Hz on the grid of budgets computed, whose neighbouring points are 0.4 and 2 Hz. That is near the mean rate of about 4 Hz that the Attwell–Laughlin budget assumes [10]. With 50 windows the information per ATP is largest near 10 Hz. Short decisions thus push the efficient rate up, away from the measured rates. For a single neuron the binding variable is the number of spikes in the decision, which points to pooling or to longer integration.

S1.5.5 What Kostal and Shinomoto found, and what changes here

Kostal and Shinomoto computed the energy–capacity trade-off of a Poisson neuron with the Attwell–Laughlin cost, together with a version for finite coding at fixed error probability [20]. They also asked whether, under some coding constraint, a continuously varied rate might help; the graded long-window codes above are one answer, with a small gain at $A = 3$ and a large one beyond the first count change. Three differences in the setup explain the differences in the results. First, their input may switch freely in continuous time, which makes the on-off code optimal; we hold the rate

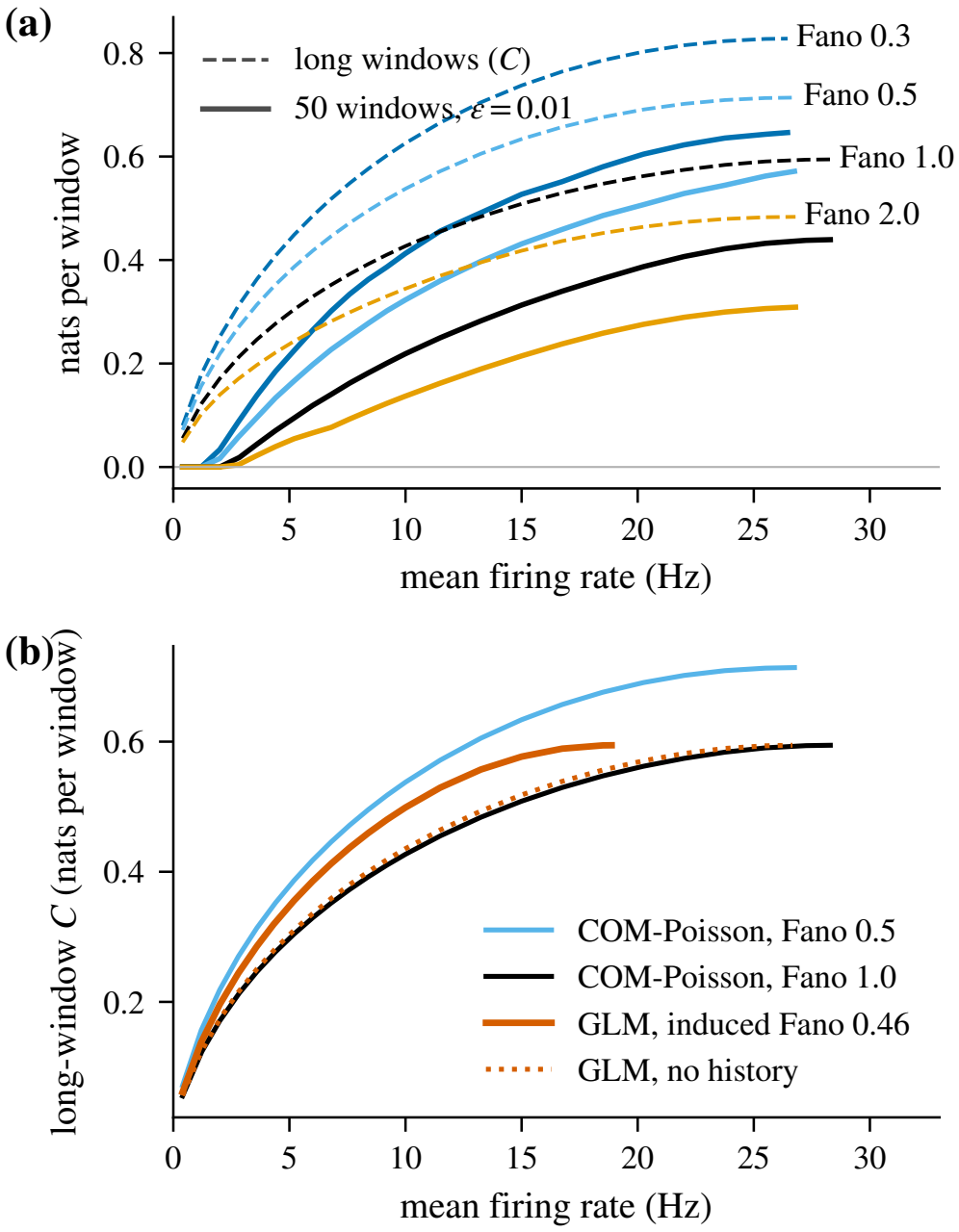


Figure 3: The energy–capacity front of a neuron. (a) COM-Poisson neurons with Fano factors 0.3 to 2: long-window capacity (dashed) and the short-window front with 50 windows at $\epsilon = 0.01$ (solid, $\log M^*$ per window). Lower Fano factor is better at every energy but one in each front (43 of 44 in short windows). (b) Long-window fronts of COM-Poisson neurons and of GLM neurons: refractory spike history lies above the Poisson neuron but below a count model of similar regularity. Windows of 50 ms, rates 0 to 60 Hz.

fixed through a window, and only this restriction allows a graded code at all. Second, they cap the peak rate at 50 Hz to keep the linear cost model and the Poisson approximation valid. At a 50 ms window that cap is $A = 2.5$, below 3.3679, so without an energy budget our long-window code is on-off there as well; their model carries an average-power constraint besides the peak, their Eq. (2), and with a budget our code can be graded at $A = 3$ (Fig. 2), so whether it is graded at $A = 2.5$ under their budget is not settled here. Third, their finite-coding version comes from an error exponent: the window grows until a target error is reached, a bandwidth-type constraint under which a low rate is paid for with time, and their efficient operating point moves to lower cost. Ours fixes the number of windows, a blocklength-type constraint that penalizes low rates instead, and the efficient point moves to higher rates. The two constraints bound the operating point from opposite sides. Our neuron family is also wider, since the COM-Poisson count covers both directions of variability and contains their Poisson neuron as one member.

S1.6 Do recorded neurons use discrete levels?

Two lines of theory predict that a single neuron's tuning curve takes a few discrete values [22, 30]. Nikitin et al. note that it is not clear whether a test with the right stimulus and counting window has been done, and Bethge et al. cite evidence for binary coding in some neurons, such as the H1 neuron of the fly. We ran a pre-registered test on six public datasets: motor cortex, area 2 and dorsomedial frontal cortex of monkeys (DANDI 000128, 000127, 000130 [31]), larval salamander retina (Dryad 10.5061/dryad.4qrfj6qm8), mouse V1 (Allen Visual Coding Neuropixels, DANDI 000021) and mouse A1 (DANDI 000986). For each tuned unit the tuning map was estimated from the trials, and the number of levels was the number of k-means clusters of that map that best predicts held-out trials. If the code were discrete, this number would stop growing once enough trials were added. Two comparators were built on each unit: a discrete one, with $N(A, \lambda)$ levels on the unit's own map and matched noise, and a continuous one, with the map itself and the same noise. Only units whose two simulations separate are scored, and a shuffle control must give one level.

The registered rule returned *mixed* in five datasets and *smooth* in the salamander retina (Table 3). The rule counts pairs whose level count is one, units without resolvable tuning, as flat, so we also report a second reading restricted to tuned pairs (level count at least two). There the level count grew with the number of trials in 57 to 85% of the tuned pairs of five datasets; in mouse A1, where the count is capped at five levels, 70% of the tuned pairs reached all five. In M1 and V1 the *mixed* verdict came from the untuned pairs, since growth among tuned pairs reached 79% and 85%; in area 2 and DMFC growth stayed below 70% even among tuned pairs. At the full trial count the level count was 1.8 to 3.3 times the discrete comparator. It was within 0.2 of the continuous comparator in two datasets and differed from it by 1.5 to 2.0 levels in the other four, in both directions. The comparators therefore calibrate the test only to about two levels. Among powered pairs with a peak count below 3.4, where the theory predicts two levels, only 2 to 15% stopped at two. The A1 recordings, with a median peak count of 0.35 per window among powered units and a predicted two levels for every one of them, are the case most favourable to the theory. The theory says what the levels would be if a neuron implemented the optimal code; these recordings do not show them. Their median Fano factors, 0.90 to 1.39, are close to one, so the regularity that brings most auditory-nerve fibres of the main text within the optimal staircase is absent here. This level-count test also differs in design from the staircase test of the main text. It also lies outside the conditions of the theory in two further ways. The window in which cortical neurons are read is not known, and the number of levels grows with the window. The stimuli were a few conditions chosen

Table 3: The level-count test on six public datasets

dataset	rule	units	levels[b]	growth	at 2
M1 (MC_Maze)	mixed	178	8.86 / 2.71 / 8.73	79%	2%
area 2	mixed	65	8.10 / 3.02 / 6.63	57%	9%
DMFC	mixed	53	5.39 / 2.96 / 7.40	59%	15%
salamander retina	smooth	381	6.55 / 3.10 / 8.43	75%	10%
mouse V1	mixed	100	6.46 / 2.67 / 7.92	85%	8%
mouse A1 (ceiling 5)	mixed	123	4.26 / 2.28 / 4.09	70%[a]	8%

rule: registered verdict; units: powered units; levels: mean level count over powered pairs, [b]data / discrete comparator / continuous comparator; growth: among tuned powered pairs (level count at least two); at 2: plateau at two levels among powered pairs with $A + \lambda < 3.4$. [a]Tuned pairs reaching all five levels.

by the experimenter, and three datasets come from motor and frontal areas that do not encode a stimulus distribution in Barlow's sense.

S1.7 Finitely many rates for Conway-Maxwell-Poisson counts

The matched test of the main text computes the number of steps for a Conway-Maxwell-Poisson count. This section shows that the optimal input of that counting channel, like that of the Poisson channel [1], has finitely many mass points, so that Proposition 1 of the main text applies to it.

For $\zeta > 0$ and $u \geq 0$ let $W_u(y) = u^y (y!)^{-\zeta} / Z(u)$ on $y = 0, 1, 2, \ldots$, with $Z(u) = \sum_y u^y (y!)^{-\zeta}$. Since $(y!)^\zeta$ grows faster than any geometric sequence, Z is an entire function, and $Z(u) \geq 1$ for $u \geq 0$ because the term $y = 0$ equals one. The mean $m(u) = uZ'(u)/Z(u)$ satisfies $u\, m'(u) = \mathrm{Var}_u(Y) > 0$ for $u > 0$, so m increases strictly. Also $m(u) = u(1 + O(u))$ as $u \to 0$, and $m(u) \to \infty$ as $u \to \infty$. Hence each rate $r > 0$ is the mean of exactly one W_u, and $u = 0$ gives the count that is always zero, as in the main text. The input interval $[\lambda, \lambda + A]$ becomes an interval $U = [u_0, u_1]$, with $u_0 = 0$ when $\lambda = 0$, and mass points in r and in u correspond one to one.

Proposition S1.7.1. *For every $\zeta > 0$, $\lambda \geq 0$ and $A > 0$, the capacity of this counting channel is attained, and every capacity-achieving input has finitely many mass points.*

Proof. Existence. For $u \in U$, $W_u(y) \leq b(y) = u_1^y (y!)^{-\zeta}$, and $\sum_y b(y) |\log b(y)| < \infty$. Every output law $q_P = \int W_u \, dP(u)$ therefore obeys $q_P \leq b$, so its entropy is a continuous function of P in the weak topology. The entropy of W_u is continuous in u on U, including $u = 0$, where it tends to zero. The mutual information is thus continuous on the compact set of input laws on U. Its maximum C is attained, and $C > 0$ because $A > 0$.

Bounds on the output law. Let P^* be capacity-achieving with output law q. P^* is not the point mass at $u = 0$, which gives no information, so some $u_a > 0$ has $p_a = P^*([u_a, u_1]) > 0$. For $u \in [u_a, u_1]$, $W_u(y) \geq u_a^y (y!)^{-\zeta} / Z(u_1)$. With the upper bound above,

$$\zeta \log y! - y \log u_1 \;\leq\; -\log q(y) \;\leq\; \zeta \log y! + y\, |\log u_a| + \log \frac{Z(u_1)}{p_a}. \tag{7}$$

Analyticity. Let $i(u) = \sum_y W_u(y) \log(W_u(y)/q(y))$, defined for every $u \geq 0$. By (7) each term of $\log W_u(y) - \log q(y) = y \log u - \zeta \log y! - \log Z(u) - \log q(y)$ has a finite mean under W_u, so for $u > 0$

$$i(u) = m(u) \log u - \log Z(u) - \frac{G(u)}{Z(u)}, \qquad G(u) = \sum_y u^y (y!)^{-\zeta} g(y), \quad g(y) = \zeta \log y! + \log q(y).$$

By (7), $|g(y)|$ grows at most like $y \log y$, so G is entire. As $Z \geq 1$ on $[0, \infty)$, the function i is real analytic on $(0, \infty)$. At $u = 0$, $m(u) \log u \to 0$ and $i(u) \to -g(0) = -\log q(0) = i(0)$, so i is continuous on $[0, \infty)$.

Optimality condition. For $u \in U$ and $0 < \varepsilon < 1$ let $P_\varepsilon = (1 - \varepsilon)P^* + \varepsilon \delta_u$. By (7), $\sum_y |W_u(y) - q(y)|\,|\log q(y)| < \infty$, which allows the derivative of the output entropy to be taken term by term. The one-sided derivative of the mutual information at $\varepsilon = 0$ is $i(u) - C$, and it cannot be positive. Hence $i \leq C$ on U, as in the argument of Smith [3]. Since $\int i \, dP^* = C$, equality holds P^*-almost everywhere, and by continuity on the whole support of P^*.

Growth. Write $M(s) = m(e^s)$ and $t_1 = \log u_1$. For $t > t_1$, the left inequality of (7) gives

$$i(e^t) \;\geq\; (t - t_1)M(t) - \log Z(e^t) \;=\; \int_{t_1}^{t} \big(M(t) - M(s)\big)\, ds - \log Z(u_1),$$

because the derivative of $\log Z(e^s)$ in s is $M(s)$. Since M increases, the right side is at least $M(t) - M(t_1 + 1) - \log Z(u_1)$ for $t \geq t_1 + 1$, so $i(u) \to \infty$ as $u \to \infty$.

Behaviour at $u = 0$. Write $i(u) = m(u) \log u + \Psi(u)$, with $\Psi = -\log Z - G/Z$ analytic at 0 and $\Psi(0) = i(0)$. Since $m(u) = u(1 + O(u))$,

$$i(u) - i(0) = u \log u + \Psi'(0)\, u + O(u^2 \log u) \qquad (u \to 0^+),$$

so $i(u) < i(0)$ for all small $u > 0$.

Conclusion. Suppose the support of P^* is infinite. It is a closed subset of the compact interval U, so it has an accumulation point u^*, and $i = C$ on the support. If $u^* > 0$, the analytic function $i - C$ has zeros accumulating in $(0, \infty)$, so $i = C$ on all of $(0, \infty)$, which contradicts the growth. If $u^* = 0$, which can happen only for $\lambda = 0$, support points $u_k \to 0$ give $i(0) = C$ by continuity, which contradicts $i(u_k) = C$ for small $u_k > 0$. Hence the support is finite. □

The value $\zeta = 1$ is the Poisson channel. The proof covers the endpoint $\lambda = 0$, which six of the 265 fibres of the matched test have. The number of steps used in the main text is the number of mass points of a numerical solution on a grid of 801 rates. The proposition shows that the exact optimum it approximates also has finitely many mass points.

S1.8 Where a new rate appears

Measured from the silent side in the Fisher coordinate, an inserted rate (odd K) appears at 0.62 of its gap for $K = 5$ to 21, and a split (even K) happens at 0.12 of the gap next to the parent rate. On the grid of Barletta and Dytso the insertion fraction is 0.61 to 0.62 for $K = 5$ to 9 and 0.57 at $K = 11$, where the grid point lies furthest above the birth. Forty-digit birth states give 0.620 to 0.623 for $K = 17$ to 21. The Gaussian channel inserts at the centre of its central gap, $\frac{1}{2}$, and splits its central level in place; Barletta and Dytso study how its support evolves as the amplitude grows [32]. A baseline carries both fractions to the Gaussian values. The insertion position falls from 0.62 through 0.61, 0.58 and 0.56 at $\lambda = 0.01$, 0.03 and 0.05 to 0.513 and 0.507 at $\lambda = 1$ and 3, and the split position falls from 0.124 to 0.06 at $\lambda = 0.05$. The negative binomial, which keeps the Poisson wall, keeps both positions (0.624 at $\rho = 300$, 0.128 at $\rho = 3000$).

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